FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Aerospace Engineering

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INTERNSHIP REPORT ON

ADDITIVE MANUFACTURING OF AN AIRFOIL FOR WIND TUNNEL TESING

by

SALMAN FARIS E T (22BTRAS038)

at

Verve Visions Pvt. Ltd.

From: 05/07/25 **To: 05/08/25**

BACHELOR OF TECHNOLOGY
in
AEROSPACE/AERONAUTICAL ENGINEERING

Signed Internship Certificate from company to be attached as second page

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ABOUT THE COMPANY

Verve Visions Private Limited is a private limited company based in Bengaluru (Vidyaranyapura, Bangalore North), Karnataka. It was incorporated on 24 January 2024 under CIN U74109KA2024PTC183905. Its registered office is at B-404, Max Spoorthi Apartments, 8th Main, Sapthagiri, Bangalore North PIN code 560097.

The company's directors are Brijesh Kumar Malekopmath and Kanta Brijesh Malekopmath, both appointed at the time of incorporation.

Its authorised share capital is ₹100,000 (₹1 lakh), and the paid-up capital is ₹10,000. According to corporate registries, its main classification is "Other Professional; Scientific and Technical Activities."

Importantly, Verve Visions positions itself as a sustainable product-design and development company. On its official website, the company states a mission to "pioneer a new era of sustainable product design, where innovation meets environmental consciousness." Its design philosophy emphasises the principles of "Reduce, Recycle, Reuse", aiming to minimise ecological footprint, use renewable or recycled materials, prioritise energy efficiency, and reduce waste in the product-life cycle.

STUDY CARRIED OUT

1. OBJECTIVES

The objective of this work is to understand the complete process involved in the design, preparation, and fabrication of wind-tunnel airfoil models, which includes:

- Selection of an appropriate airfoil based on aerodynamic performance characteristics
- CAD modeling, geometric refinement, and integration of pressure measurement features
- Preparation of the digital model for additive manufacturing
- Fabrication and post-processing of the 3D-printed airfoil for wind-tunnel testing

2. INTRODUCTION

Wind-tunnel testing remains one of the most reliable and widely used methods for evaluating aerodynamic behavior. To achieve accurate and repeatable results, it is essential to use precisely manufactured test models with high geometric fidelity. With advancements in digital engineering, Computer-Aided Design (CAD) and Additive Manufacturing (3D printing) have become highly effective tools for producing detailed and accurate aerodynamic models.

In this work, the Michael Selig S1223 airfoil, known for its high-lift characteristics and performance in low Reynolds number applications, was selected for fabrication. A detailed 3D model of the airfoil was created, incorporating 22 pressure vents along the chord for capturing pressure distribution during wind-tunnel testing. The use of 3D printing allowed for the production of a smooth, dimensionally accurate model while significantly reducing fabrication time compared to conventional methods.

This methodology highlights the integration of modern CAD workflows and additive manufacturing techniques in the development of aerodynamic test articles, demonstrating how digital tools can enhance precision, efficiency, and overall reliability in experimental aerodynamics.

3. DESIGN METHODOLOGY

The methodology adopted for the design, preparation, and fabrication of the wind-tunnel airfoil model involved the following systematic steps:

3.1 Selection of Airfoil Profile

A suitable airfoil was identified based on aerodynamic performance requirements. For this study, the Michael Selig S1223 airfoil was chosen after evaluating its aerodynamic characteristics, flow behaviour, and suitability for low Reynolds number operations.

The S1223 airfoil is known for its exceptionally high maximum lift coefficient, often exceeding values achieved by conventional airfoils of similar thickness. This superior lift performance is attributed to its highly cambered profile, which enhances suction on the upper surface even at relatively low angles of attack. Compared to standard NACA 4-series or Clark-Y type airfoils, the S1223 demonstrates a much steeper lift-curve slope, enabling efficient lift generation in low-speed, low-Reynolds-number testing environments.

A key distinguishing feature of the S1223 is its sharply tapered trailing edge, which minimizes flow separation and reduces trailing-edge thickness effects that can distort aerodynamic measurements. This sharp trailing edge ensures cleaner wake formation, leading to more accurate pressure distribution data especially important when integrating multiple pressure taps along the chord.

The airfoil also exhibits a relatively thick leading-edge region, allowing it to maintain laminar flow over a significant portion of the chord. This promotes delayed boundary-layer transition and enhances aerodynamic efficiency. Additionally, the high camber-to-thickness ratio and smooth curvature distribution make the S1223 uniquely suitable for experimental setups that focus on lift enhancement and low-speed performance analysis.

The combination of high lift capability, sharp trailing edge, extended laminar flow, and predictable stall characteristics makes the S1223 airfoil an ideal choice for wind-tunnel testing. Its

unique geometry ensures that detailed pressure measurements such as those obtained through the 22 integrated pressure vents reflect accurate and meaningful aerodynamic behavior.

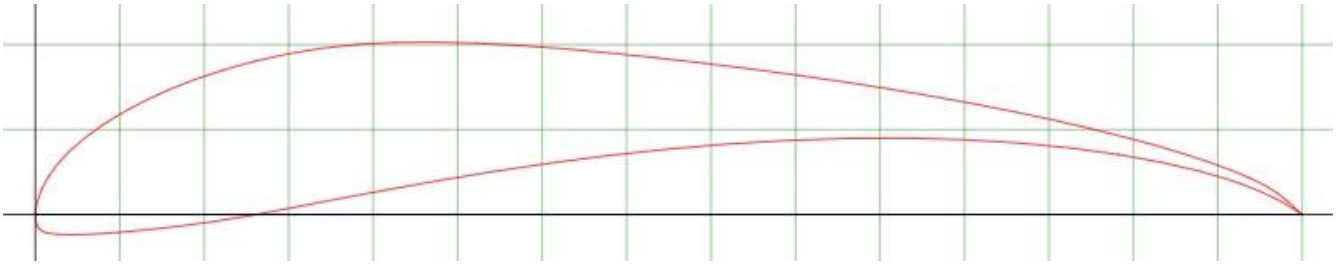


Fig1. Michael Selig S1223 airfoil

3.2 CAD Modelling and Geometric Refinement

The selected S1223 airfoil coordinates were imported into CAD software to initiate the digital modelling process. These coordinates, typically obtained in .dat format, were plotted as discrete points and then fitted using spline or curve-generation tools to recreate the exact aerodynamic profile. Ensuring accurate curve fitting at this stage was essential to maintain the original aerodynamic intent of the airfoil.

Once the 2D profile was validated, a high-fidelity 3D model of the airfoil was developed by extruding the section along the span. Particular attention was given to maintaining surface smoothness and geometric continuity, as even minor irregularities in the model can significantly influence the airflow during wind-tunnel testing.

The chord length, and camber line were precisely replicated based on the standard S1223 specifications. Refinement operations were performed to eliminate any artifacts generated during point-to-curve conversion, ensuring a clean aerodynamic surface. Additional geometric features such as the sharp trailing edge, pressure-tap locations, and internal passage provisions were integrated while ensuring that they did not distort the aerodynamic shape.

To prepare the model for fabrication, checks were conducted for solid body integrity and alignment of pressure vents. The final CAD model represented a dimensionally accurate and manufacturable version of the S1223 airfoil, forming the foundation for 3D printing and subsequent wind-tunnel experimentation.

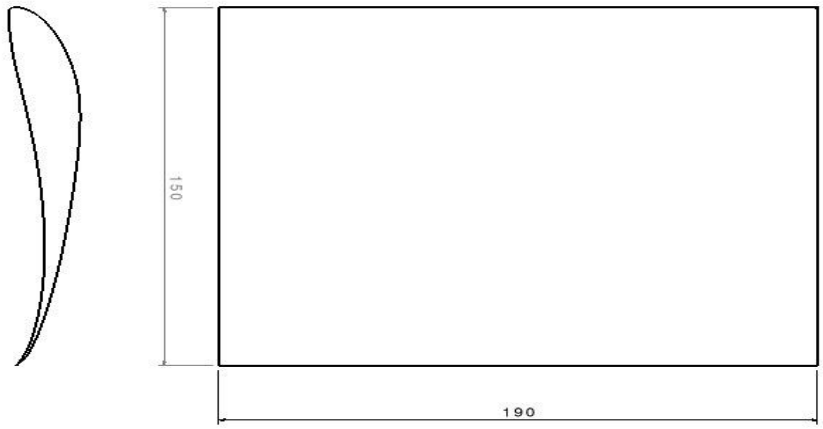


Fig 2. 2-D sketch

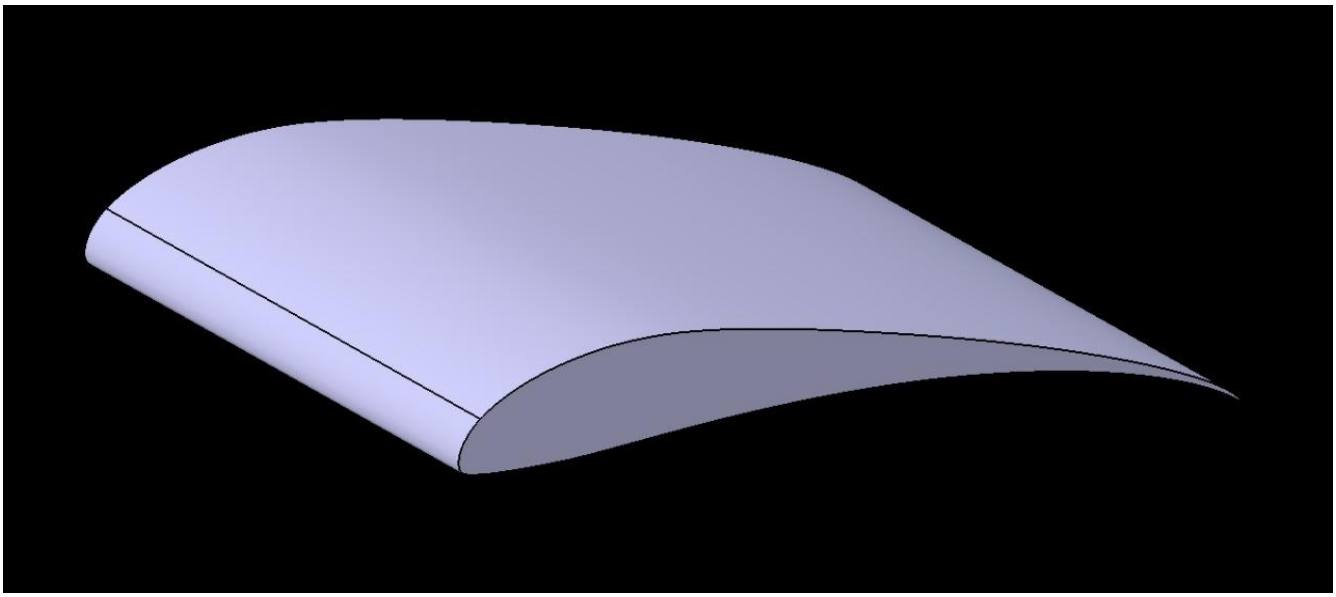


Fig 3. 3D model

3.3 Pressure Vent Placement

To obtain a precise and high-resolution pressure distribution across the airfoil surface, 22 pressure measurement vents were incorporated into the model. These vents were distributed along the chord at locations chosen to capture key aerodynamic phenomena. A significant portion of these 12 vents, more than half of the total were concentrated within the first 30% of the airfoil chord. This region contains the leading-edge suction peak and the steepest pressure gradient, where rapid pressure variations occur due to flow acceleration over the highly cambered S1223 profile.

Capturing this zone in finer detail is essential for accurately understanding the lift characteristics and boundary-layer behaviour.

Each pressure vent was designed and oriented normal to the local airfoil surface to ensure undistorted static pressure readings. Normal alignment prevents the introduction of dynamic pressure effects or flow-angle biases, which could occur if the vent were inclined relative to the external flow direction. This alignment is critical for generating reliable wind-tunnel data.

The vents were modelled as small-diameter internal channels extending from their surface openings to a common outlet region along the span. The routing of these passages required careful planning to ensure they did not intersect, weaken the structure, or distort the external aerodynamic contour. The openings were created flush with the airfoil surface to maintain smooth airflow and prevent localized disturbances.

Additionally, the CAD model was refined to ensure that the presence of pressure ports did not compromise the structural integrity or manufacturability of the 3D-printed component. Clearances were provided for tube insertion, and the internal walls around the channels were checked for adequate strength.

The pressure-vent integration strategy ensured a high-fidelity, high-resolution measurement setup, enabling accurate capture of the pressure distribution and aerodynamic behaviour of the S1223 airfoil during wind-tunnel testing.

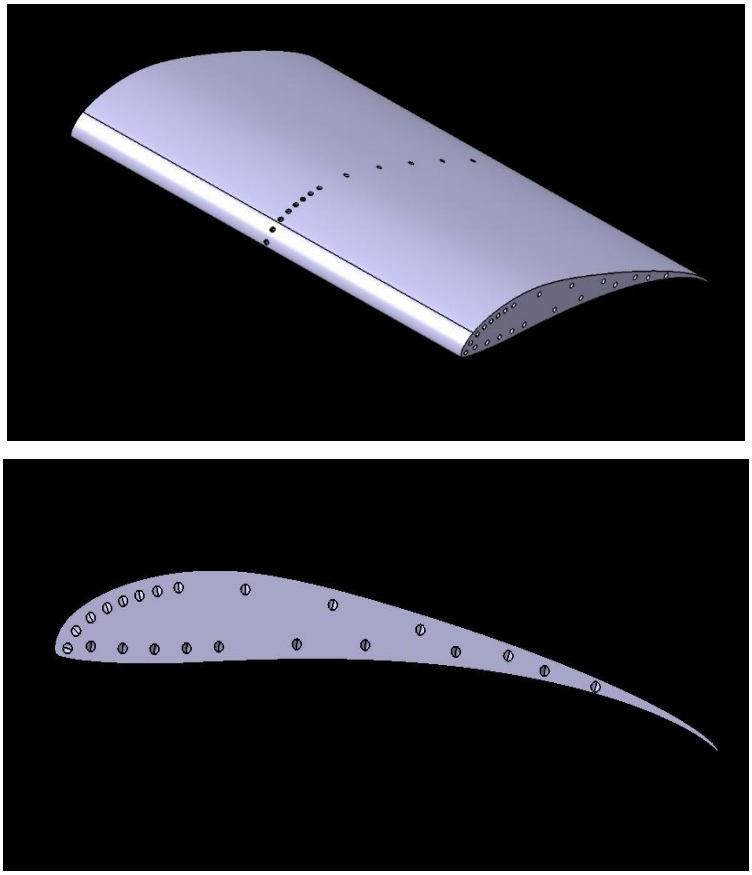


Fig 4. Pressure vent placement

3.4 Preparation for Additive Manufacturing

The finalized CAD model was prepared for fabrication using the Ultimaker Cura slicing software. During this stage, various parameters such as part orientation, support generation, infill density, and layer height were evaluated to ensure that the aerodynamic surfaces were preserved with maximum fidelity. The model was exported in STL format, maintaining the exact geometric profile and dimensional accuracy required for wind-tunnel testing.

During slicing, it was observed that the S1223 airfoil's extremely sharp trailing edge could not be accurately reproduced by the Fused Deposition Modelling (FDM) 3D printer being used. FDM technology has inherent limitations in printing very thin, zero-thickness edges due to nozzle diameter constraints, material flow, and cooling characteristics. Attempting to print the original trailing edge resulted in deformation, incomplete extrusion, and inconsistent geometry.

To overcome this limitation while still achieving the required aerodynamic accuracy, a design modification strategy was adopted. The trailing edge section was intentionally removed (truncated)

in the CAD model prior to slicing. This ensured clean and reliable printing of the main airfoil body without introducing surface defects or print failures.

Following 3D printing, the missing trailing edge was reconstructed manually using a thin foam sheet that could be sanded and shaped to achieve the precise sharpness characteristic of the S1223 profile. The foam material allowed the creation of a razor-sharp trailing edge that the FDM printer could not produce, thereby restoring the correct aerodynamic geometry for wind-tunnel testing. The foam extension was bonded to the printed model with care to ensure proper alignment, surface continuity, and smooth airflow transition.

This hybrid approach combining additive manufacturing with manual trailing-edge fabrication allowed the team to retain the manufacturing convenience of FDM printing while still achieving the high-fidelity aerodynamic surface required for accurate experimental analysis.

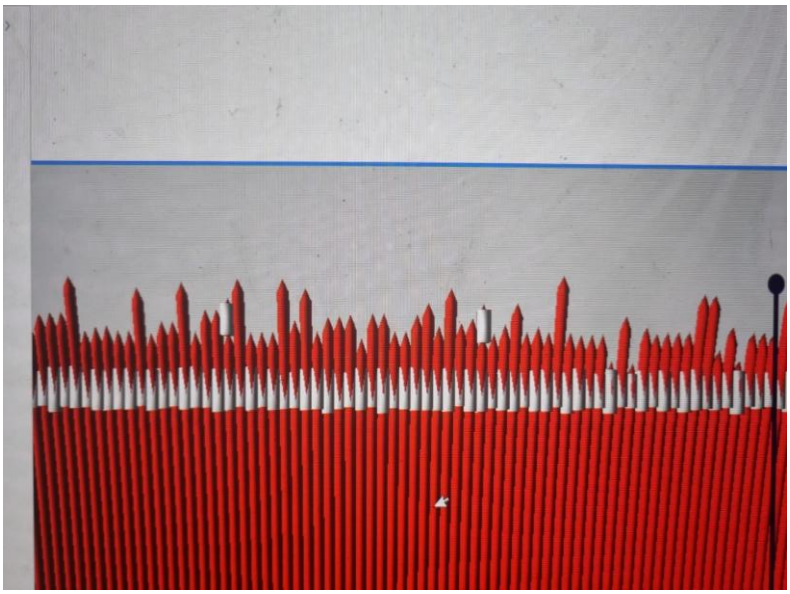


Fig 5. This picture depicts the region of the trailing edge that could not be printed using a 3D printer due to its sharp edge.

3.5 Additive Manufacturing (3D Printing)

The airfoil model was fabricated using Fused Deposition Modelling (FDM) technology on the Bambu Lab A1 3D printer. The system offers a build volume of 256 × 256 × 256 mm, a maximum toolhead speed of 500 mm/s, and a hot-end temperature capability of 300°C, making it highly suitable for producing precise aerodynamic components. Its all-metal hot end, automatic

calibration features, and closed-loop cooling system contributed to consistent extrusion and high dimensional accuracy.

The printing material selected was PLA (Polylactic Acid), which is one of the recommended filaments for the Bambu Lab A1. PLA was chosen due to its ease of printing, dimensional stability, low warping tendency, and ability to reproduce fine geometric details, all of which are crucial for preserving the aerodynamic profile of the S1223 airfoil.

A 0.4 mm nozzle was used to provide adequate resolution for the curved surfaces and thickness variations of the airfoil. Printing parameters such as layer height, print speed, and infill density were optimized to achieve a balance between surface smoothness and structural integrity. A low layer height was applied to minimize surface roughness and ensure clean aerodynamic flow during wind-tunnel testing, while a moderate infill percentage kept the model lightweight yet sufficiently rigid.

The PEI-coated build plate ensured strong adhesion throughout the print, reducing any risk of warping or surface deviation. The printer's automated features including pressure advance calibration, belt tension monitoring, and a filament run-out sensor helped maintain a stable and defect-free manufacturing process. Slicing was performed using Bambu Studio, ensuring proper tool-path generation and accurate reproduction of the airfoil geometry.

The use of PLA material in combination with the advanced capabilities of the Bambu Lab A1 resulted in a high-precision, aerodynamically accurate, and structurally stable airfoil model suitable for wind-tunnel testing.

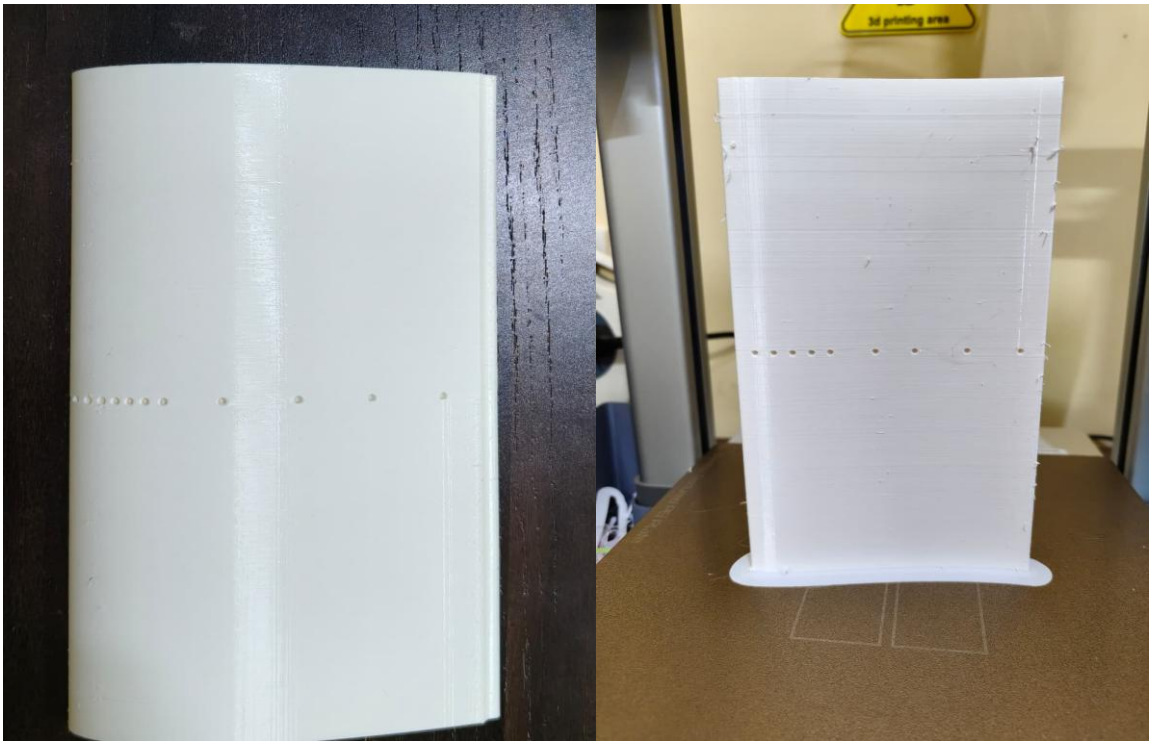


Fig 6. Final airfoil model after 3-D printing

3.6 Post-Processing and Quality Verification

After the completion of the 3D printing process, the airfoil model underwent systematic post-processing to ensure aerodynamic accuracy and readiness for wind-tunnel testing. Initial steps included the removal of all support structures, cleaning of surface irregularities, and clearing of the 22 pressure vents to ensure unobstructed paths for pressure tubes and accurate sensor readings. Each vent was inspected to verify that its opening remained normal to the airfoil surface and free from printing artifacts.

Due to FDM printing limitations in reproducing the sharp trailing edge of the S1223 profile, the trailing edge region had been intentionally truncated during the design stage. In post-processing, the missing trailing-edge portion was manually reconstructed using a thin foam sheet, which enabled the creation of a razor-sharp, aerodynamically accurate tip that FDM technology could not produce. The foam piece was precisely cut, bonded to the printed airfoil, and aligned carefully to ensure continuity with the original profile.

To achieve a smooth aerodynamic surface essential for reliable wind-tunnel performance, the model was sanded using fine-grit sandpaper. This step helped eliminate layer lines from the FDM printing process, improved surface uniformity, and ensured that the airflow over the model remained undisturbed. Special attention was given to the leading-edge region and the newly added trailing-edge surface to preserve their geometric accuracy.

Dimensional verification was performed using calipers and profile templates to confirm that the printed and finished model closely matched the original CAD geometry. Checks included chord length, thickness distribution, pressure-port spacing, and alignment of the foam trailing-edge extension. Only after confirming compliance with the design tolerances was the model approved for wind-tunnel integration

WORK RESPONSIBILITIES DURING THE INTERNSHIP

During the internship, I was involved in the complete workflow of designing and fabricating a wind-tunnel airfoil model. I began by importing the S1223 airfoil coordinates into CATIA and creating a high-fidelity 3D model, including internal channels for pressure measurement vents. The CAD model was refined to maintain geometric accuracy and aerodynamic continuity, with modifications such as truncating the trailing edge to suit FDM printing limitations.

The model was then prepared for additive manufacturing using Ultimaker Cura, where I optimized layer height, infill, support structures, and print orientation. The airfoil was 3D printed in PLA, and I oversaw the printing process to ensure dimensional accuracy and surface quality. Post-processing involved removing supports, clearing pressure vents, sanding, and manually reconstructing the trailing edge using a foam sheet. Finally, I conducted dimensional verification and prepared the model for wind-tunnel testing, ensuring accurate pressure instrumentation and aerodynamic readiness.

SKILLS LEARNED/ SKILLS ENHANCED

1. CAD Modelling in CATIA (Airfoil Geometry and Model Preparation):

During the internship, I strengthened my CAD modelling skills using CATIA, particularly in developing accurate airfoil geometries from coordinate data. I learned how to refine profiles, ensure smooth geometric continuity, and construct the internal features required for pressure distribution studies. This included preparing CAD models with proper vent routing, thickness adjustments, and trailing-edge modifications suitable for additive manufacturing.

2. Slicing and Print Preparation Using Ultimaker Cura:

I gained hands-on experience using Ultimaker Cura for slicing and preparing models for FDM 3D printing. This included selecting optimal orientations, generating supports, tuning print parameters such as layer height and infill, and ensuring that the exported G-code was suitable for fabricating a smooth and dimensionally accurate aerodynamic model.

3. Additive Manufacturing (FDM 3D Printing) Using PLA:

I developed practical skills in FDM-based additive manufacturing, including operation of the printer, loading PLA filament, troubleshooting print issues, and verifying the quality of printed components. I also learned to adapt the CAD model to account for limitations of FDM, such as handling the trailing edge by replacing it with a foam sheet for accuracy and smoothness.

4. Post-Processing and Aerodynamic Model Finishing:

I learned fabrication techniques required after printing, such as support removal, vent cleaning, sanding, and surface finishing to achieve the smooth aerodynamic surface required for wind-tunnel testing. This included manually constructing and attaching the foam-sheet trailing edge and performing quality checks to ensure the printed model matched the CAD dimensions

CHALLENGES FACED AND SOLUTIONS

Challenge 1: Inability of FDM Printer to Produce Extremely Sharp Trailing Edge

The S1223 airfoil has a razor-sharp trailing edge that exceeded the physical limits of the FDM printing process.

Solution:

The trailing edge was intentionally truncated in the CAD model. After printing, the sharp edge was reconstructed manually using foam sheet, enabling us to achieve the required aerodynamic geometry.

Challenge 2: Maintaining Accurate Pressure-Vent Geometry

With 22 pressure vents routed internally, ensuring clean channels and precise surface openings was difficult.

Solution:

Each vent was manually cleaned, drilled, and verified after printing. CAD alignment ensured the openings remained normal to the surface for accurate pressure measurements.

Challenge 3: Achieving Smooth Aerodynamic Surface Finish

Layer lines and print artifacts could affect airflow during wind-tunnel testing.

Solution:

Fine-grit sandpaper was used to finish the airfoil surface, improving smoothness and reducing flow disturbances

CONCLUSION

This internship provided valuable exposure to the complete workflow of designing and fabricating a wind-tunnel airfoil model using CAD and additive manufacturing. I learned how to generate accurate 3D models from airfoil data, integrate pressure-measurement vents, and prepare the geometry for FDM printing while considering manufacturing constraints. The experience of slicing, printing with PLA, resolving limitations such as the sharp trailing edge, and performing post-processing enhanced my practical understanding of 3D-printing technology.

I also developed skills in surface refinement, pressure-vent calibration, and ensuring dimensional accuracy through verification against CAD specifications. Beyond technical modeling and fabrication, this project improved my understanding of how aerodynamic models are prepared for experimental testing and how pressure distribution is evaluated in practice. Overall, the internship strengthened my proficiency in CAD modelling, additive manufacturing, wind-tunnel model preparation, and practical problem-solving in an aerospace testing environment.